Lecture 9 - Midterm Review

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One long example, weaving in review along the way:
Block diagrams $\rightarrow$ Transfer Functions $\rightarrow$ Control design and stability $\rightarrow$ S.S response $\rightarrow$ Sys type.

Block Diagrams

1. Consider the following block diagram with output Y and 2 external inputs:
   - Reference R
   - Disturbance W
2. Let's simplify, design controller, and make sure system stable for plant $G = \frac{1}{s+10}$

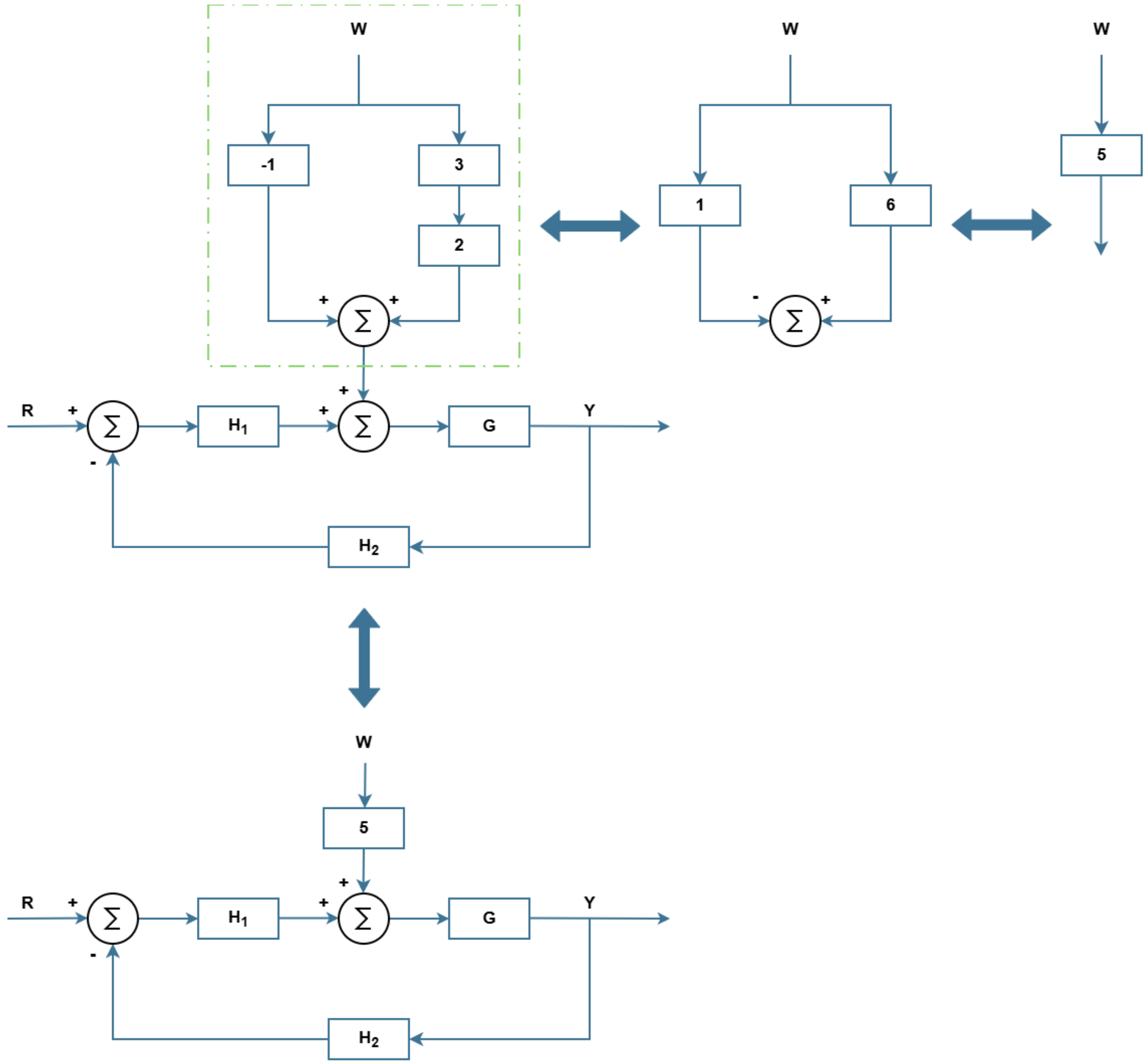

Figure 9.1: Block diagram simplification example.

Let's find transfer functions.
$G_{RY}: \ W = 0$

$$
\begin{aligned}
&Y = G \cdot H_1 \left[R - (H_2)Y\right] \\
&Y = GH_1R - GH_1(H_2)Y \\
&Y\left[1 + GH_1(H_2)\right] = GH_1R \\
&G_{RY} = \frac{Y}{R} = \frac{GH_1}{1 + GH_1(H_2)}
\end{aligned}
$$

Note:, the T.F. for G_{RY} that we often see, $G_{RY} = \dfrac{GH}{1+GH}$, is associated to a specific block diagram we often use.

G_{WY}: R $= 0$

$$
\begin{aligned}
&Y = G \cdot [5W + H_1(-1)(H_2)Y] \\
&Y = 5GW - GH_1(H_2)Y \\
&Y\,[1 + GH_1(H_2)] = 5GW \\
&G_{WY} = \frac{Y}{W} = \frac{5G}{1 + GH_1(H_2)}
\end{aligned}
$$

Note: Denominator is same as for G_{RY}. We use denom to asses stability.

Let's plug in values in our blocks and evaluate how a PI controller performs.

Let's use 1st order plant:

$$G = \frac{1}{s+10}, \qquad \text{Use}: \ H_2 = 1,$$

And design a PI controller:

$$H_1 = k_p + \frac{k_I}{s}$$

$\rightarrow$ Check stability, steady state response, and system type.

1. Stability
   - Closed loop stability determined by characteristic equation (denominator).

     $$\frac{\sim}{1 + GH_1(H_2)} \quad \leftarrow \quad \text{for our system}$$

   - Use Routh Hurwitz criterion to check coefficients of polynomial.

     $$
     \begin{aligned}
     &\underline{\text{1st order:}} \quad \frac{1}{s+a_1} \rightarrow \text{stable if } a_1 > 0 \\
     &\underline{\text{2nd order:}} \quad \frac{1}{s^2 + a_1 s + a_2} \rightarrow \text{stable if } a_1, a_2 > 0 \\
     &\underline{\text{3rd order:}} \quad \frac{1}{s^3 + a_1 s^2 + a_2 s + a_3} \rightarrow \text{stable if: } \begin{cases} a_1 > 0,\ a_2 > 0,\ a_3 > 0 \\ a_1 \cdot a_2 > a_3 \end{cases}
     \end{aligned}
     $$

     $\rightarrow$ Our system

     $$\frac{1}{1 + GH_1(H_2)} = \frac{1}{1 + \frac{(sk_p + k_I)}{s(s+10)} 1} = \frac{1}{s(s+10) + sk_p + k_I} = \frac{1}{s^2 + (10 + k_p)s + k_I}$$

     $$
     \begin{aligned}
     \therefore \quad &\text{For stability, } 10 + k_p > 0 \rightarrow k_p > -10 \\
     &k_I > 0
     \end{aligned}
     $$

   - Can we also specify performance characteristics (e.g. prevent oscillation) by placing poles?
     - Yes: 2 poles and 2 DoF, can set ω_n and ζ.
       Recall: $\frac{1}{s^2 + 2\zeta\omega_n + \omega_n^2}$ is the standard 2nd order form.
       $\rightarrow$ e.g. choose $\omega_n = 6 \rightarrow k_I = \omega_n^2 = 36$
       Choose critically damped:
       Choose $\zeta = 1 \rightarrow 10 + k_p = 2\omega_n\zeta = 2\omega_n = 12 \rightarrow k_p = 2$

2. Steady state response

- Closed loop output:

$$Y(s) = G_{RY}(s)R(s) + G_{WY}W(s) \quad \leftarrow \text{Our system only has 2 external inputs: } R \text{ \& } W$$

- Steady state output:

$$\lim_{t\to\infty} y(t) = \lim_{s\to\ 0} sY(s) = \lim_{s\to\ 0} s\left[G_{RY}(s)R(s) + G_{WY}(s)W(s)\right]$$

Example: What's the steady state output for a step input $r(t) = 5$ (reference = 5 for all time) and a step disturbance $w(t) = 1$ (constant disturbance)?

Let's use our values for k_p and k_I: $k_p = 2,\ k_I = 6$

First find $G_{RY}, G_{WY}, R(s), W(s)$

$$G_{RY} = \frac{GH_1}{1+GH_1(1)} = \frac{\frac{sk_p+k_I}{s(s+10)}}{1+\frac{sk_p+k_I}{s(s+10)}} = \frac{sk_p+k_I}{s(s+10)+sk_p+k_I} = \frac{sk_p+k_I}{s^2+(10+k_p)s+k_I} = \frac{2s+36}{s^2+12s+36}$$

$$G_{WY} = \frac{5G}{1+GH_1(1)} = \frac{\frac{5}{s+10}}{1+\frac{sk_p+k_I}{s(s+10)}} = \frac{5s}{s(s+10)+sk_p+k_I} = \frac{5s}{s^2+(10+k_p)s+k_I} = \frac{5s}{s^2+12s+36}$$

$$R(s) = \frac{5}{s}, \quad W(s) = \frac{1}{s}$$

$$\lim_{t\to\infty} y(t) = \lim_{s\to\ 0} \cancel{s}\left[\frac{(2s+36)}{s^2+12s+36}\cdot\frac{5}{\cancel{s}} + \frac{5s}{s^2+12s+36}\cdot\frac{1}{\cancel{s}}\right]$$

$$= \frac{36}{36}\cdot 5 + \frac{0}{36} = \boxed{5}$$

$\rightarrow$ Good! This is what we wanted. And it tells us that our system type is large enough to both: track step R with 0 error, and reject step W

3. System Type

Now that we have Y(s), it's easy to find E(s) (the closed loop error) and evaluate system type.

- <u>Closed loop error</u>

$E(s) = R(s) - Y(s)$ (This is the closed loop Y)

$E(s) = R(s) - [G_{RY}(s)R(s) + G_{WY}(s)W(s)]$

$E(s) = (1 - G_{RY}(s))R(s) - G_{WY}(s)W(s) = G_{RE}(s)R(s) + G_{WE}(s)W(s)$

Recall, $(1 - G_{RY}(s)) = S(s) = \dfrac{1}{1+GH}$ for our familiar block diagram, but not always!

- <u>Steady state error</u>

$$\lim_{t\to\infty} e(t) = \lim_{s\to\ 0} sE(s)$$

We could now find e(t) like we did for y(t) by evaluating G_{RE}, G_{WE} and applying some r(t), w(t). But when we discuss system type, we've been looking at the external inputs (r, w, etc) separately so we understand the constraints on each signal independently.

- System Type
  - w.r.t Reference R $\rightarrow$ tells us about tracking performance
  - w.r.t Disturbance W $\rightarrow$ tells us about disturbance rejection

– $\rightarrow$ Sys type "K" tells us that steady-state error is bounded for an input polynomial of order K.

Example: <u>System type of our system w.r.t R:</u>

we had: $G_{RY} = \dfrac{GH_1}{1+GH_1(H_2)}, \quad H_2 = 1$

$\rightarrow G_{RY} = \dfrac{GH_1}{1+GH_1}$

$G_{RE} = 1 - G_{RY} = \dfrac{1}{1+GH_1} = S(s) \leftarrow$ Sensitivity function

If S(s) is stable, then:

$$r(t) = \frac{t^k}{k!} \rightarrow \underset{t\to\infty}{e(t)} = \lim_{s\to 0} \frac{1}{s^k} S(s)$$

Step: $r(t) = 1 \rightarrow \underset{t\to\infty}{e(t)} = \lim_{s\to 0} S(s)$

Ramp: $r(t) = t \ldots = \lim_{s\to 0} \dfrac{1}{s} S(s)$

Parabola: $r(t) = \dfrac{t^2}{2} \ldots = \lim_{s\to 0} \dfrac{1}{s^2} S(s)$

To track higher order signals, we'll have to design our controller or rely on our plant to cancel the $\frac{1}{s}$ terms.

Our system:

$$S(s) = \frac{1}{1+GH_1} = \frac{1}{1+\frac{sk_p+k_I}{s(s+10)}} = \frac{s(s+10)}{s(s+10)+sk_p+k_I} = \frac{s(s+10)}{s^2+(10+k_p)s+k_I}$$

$\therefore$ System Type $= 1 \leftarrow$ Can track step reference with 0 error and ramp reference with bounded error ($\frac{10}{k_I}$), (parabolic $\rightarrow \infty$ error).